



main.c...

Output



```

1  #include <iostream>
2  #include <iomanip>
3  #include <string>
4  using namespace std;
5  int main() {
6      // 3D array:
7          vitals[patient][vitalType][hour]
8      // patient: 0-2 -> Patient 1, 2, 3
9      // vitalType: 0=HeartRate(bpm) 1
10         =SystolicBP(mmHg) 2
11         =DiastolicBP(mmHg) 3
12         =Temperature(C)
13     // hour: 0-5 -> hour of the 6
14         -hour shift
15     float vitals[3][4][6] = {
16         // ----- Patient
17         1 -----
18         {
19         {72, 75, 78, 80, 76, 74},
20             // Heart Rate
21         {118, 120, 122, 121, 119, 118},
22             // Systolic BP
23         {78, 80, 79, 78, 77, 76},
24             // Diastolic BP
25         {36.8, 36.9, 37.0, 37.1, 36.8, 36.8}
26             // Temperature
27         ,

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```
16  {36.8, 36.9, 37.0, 37.1, 36.9, 36.8}
    // Temperature
17  },
18  // ----- Patient 2
    -----
19  {
20      {88, 95, 105, 112, 108, 99},
21      {130, 135, 142, 145, 140, 133},
22      {85, 88, 92, 95, 90, 87},
23      {37.2, 37.4, 37.6, 37.8, 37.5, 37.3}
24      },
25      // ----- Patient 3
          -----
26      {
27          {65, 62, 58, 55, 60, 63},
28          {105, 102, 98, 95, 100, 103},
29          {70, 68, 65, 62, 66, 69},
30          {35.8, 35.9, 36.0, 36.1, 36.0, 35.9}
31      }
32  };
33  int p, v, h; // p = patient, v =
                vital type, h = hour
34  string vitalNames[4] = {"Heart
                           Rate", "Systolic BP",
                           "Diastolic BP",
                           "Temperature"};
35  // Establish fixed decimal point
    format with 1 decimal
    precision
```

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```
35      // Establish fixed decimal point
      format with 1 decimal place
      precision
36      cout << fixed << setprecision(1);
37      // ----- TODO 1: Display the
      full dataset -----
38      // This outputs exactly 72
      labeled lines mapping Patient
      , Vital Type, and Hour
39      cout << "--- TODO 1: Labeled
      Printout of All 72 Readings
      ---" << endl;
40  for (p = 0; p < 3; p++) {
41  for (v = 0; v < 4; v++) {
42  for (h = 0; h < 6; h++) {
43      cout << "Patient " << (p + 1)
44      << " | Vital: " << left << setw(15)
      << vitalNames[v]
45      << " | Hour " << h << ": ";
46  if (v == 3) {
47      cout << vitals[p][v][h] << " C" <<
      endl;
48  } else {
49      cout << (int)vitals[p][v][h] <<
      endl;
50  }
51  }
52  }
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```

52     }
53 }
54     cout << endl;
55     // ----- TODO 2: Average of
        each vital sign, per patient
        -----
56     cout << "--- TODO 2: Average of
        Each Vital Sign, Per Patient
        ---" << endl;
57     for (p = 0; p < 3; p++) {
58         cout << "Patient " << (p + 1) <<
            " Averages:" << endl;
59         for (v = 0; v < 4; v++) {
60             float sum = 0.0;
61             for (h = 0; h < 6; h++) {
62                 sum += vitals[p][v][h];
63             }
64             float average = sum / 6.0;
65             if (v == 3) {
66                 // Formatting temperature averages
                    with 2 decimal places for
                    clinical precision
67                 cout << " " << left << setw(15) <<
                    vitalNames[v] << ": "
68                 << setprecision(2) << average << " C"
                    << endl;
69             } else {
70                 cout << " " << left << setw(
                    vitalNames[v] << ": "

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```
70  cout << " " << left << setw(15) <<
    vitalNames[v] << ": "
71  << setprecision(1) << average <<
    endl;
72  }
73  }
74  cout << endl;
75  }
76  // Reset precision to 1 decimal
    place
77  cout << fixed << setprecision(1);
78  // ----- TODO 3: Highest
    Heart Rate reading overall
    -----
79  cout << "--- TODO 3: Highest
    Heart Rate Reading Overall
    ---" << endl;
80  float maxHeartRate =
    vitals[0][0][0];
81  int maxPat = 0;
82  int maxHour = 0;
83  for (p = 0; p < 3; p++) {
84  for (h = 0; h < 6; h++) {
85  if (vitals[p][0][h] >
    maxHeartRate) {
86  maxHeartRate = vitals[p][0][h];
87  maxPat = p;
88  maxHour = h;
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```

88         maxHeartRate = maxHeartRate > val ? val : maxHeartRate;
89     }
90 }
91 }
92     cout << "Highest Heart Rate: " <<
        (int)maxHeartRate << " bpm
        (Patient " << (maxPat + 1)
93     << " at Hour " << maxHour <<
        ")" << endl;
94     cout << endl;
95     // ----- TODO 4: Flag
        abnormal vitals -----
96     cout << "--- TODO 4: Flag
        Abnormal Vitals ---" << endl;
97     // Ranges: HR 60-100, SBP 90-140,
        DBP 60-90, Temp 36.0-37.5
98     for (p = 0; p < 3; p++) {
99     for (v = 0; v < 4; v++) {
100    for (h = 0; h < 6; h++) {
101        float val = vitals[p][v][h];
102        bool isAbnormal = false;
103    if (v == 0) { // Heart Rate
104        if (val < 60 || val > 100)
            isAbnormal = true;
105    }
106    else if (v == 1) { // Systolic BP
107        if (val < 90 || val > 140) isAbnormal
            = true;
108    }

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```
108     }
109 ▾ else if (v == 2) { // Diastolic BP
110     if (val < 60 || val > 90) isAbnormal
        = true;
111     }
112 ▾ else if (v == 3) { // Temperature
113     if (val < 36.0 || val > 37.5)
        isAbnormal = true;
114     }
115 ▾ if (isAbnormal) {
116     cout << "[ALERT] Patient " << (p + 1
        ) << " | Vital: " << left <<
        setw(13) << vitalNames[v]
117     << " | Hour " << h << " | Reading:
        ";
118 ▾ if (v == 3) {
119     cout << val << " C" << endl;
120 ▾ } else {
121     cout << (int)val << endl;
122     }
123     }
124     }
125     }
126 }
127     cout << endl;
128     return 0;
129 }
130 |
```

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```
--- TODO 1: Labeled Printout of All 72  
Readings ---
```

```
Patient 1 | Vital: Heart Rate      | Hour 0:  
72
```

```
Patient 1 | Vital: Heart Rate      | Hour 1:  
75
```

```
Patient 1 | Vital: Heart Rate      | Hour 2:  
78
```

```
Patient 1 | Vital: Heart Rate      | Hour 3:  
80
```

```
Patient 1 | Vital: Heart Rate      | Hour 4:  
76
```

```
Patient 1 | Vital: Heart Rate      | Hour 5:  
74
```

```
Patient 1 | Vital: Systolic BP     | Hour 0:  
118
```

```
Patient 1 | Vital: Systolic BP     | Hour 1:  
120
```

```
Patient 1 | Vital: Systolic BP     | Hour 2:  
122
```

```
Patient 1 | Vital: Systolic BP     | Hour 3:  
121
```

```
Patient 1 | Vital: Systolic BP     | Hour 4:  
119
```

```
Patient 1 | Vital: Systolic BP     | Hour 5:  
118
```

```
Patient 1 | Vital: Diastolic BP    | Hour 0:  
78
```





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```
Patient 1 | Vital: Diastolic BP | Hour 1:
80
Patient 1 | Vital: Diastolic BP | Hour 2:
79
Patient 1 | Vital: Diastolic BP | Hour 3:
78
Patient 1 | Vital: Diastolic BP | Hour 4:
77
Patient 1 | Vital: Diastolic BP | Hour 5:
76
Patient 1 | Vital: Temperature | Hour 0:
36.8 C
Patient 1 | Vital: Temperature | Hour 1:
36.9 C
Patient 1 | Vital: Temperature | Hour 2:
37.0 C
Patient 1 | Vital: Temperature | Hour 3:
37.1 C
Patient 1 | Vital: Temperature | Hour 4:
36.9 C
Patient 1 | Vital: Temperature | Hour 5:
36.8 C
Patient 2 | Vital: Heart Rate | Hour 0:
88
Patient 2 | Vital: Heart Rate | Hour 1:
95
Patient 2 | Vital: Heart Rate | Hour 2:
105
```





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```
Patient 2 | Vital: Heart Rate | Hour 3:
112
Patient 2 | Vital: Heart Rate | Hour 4:
108
Patient 2 | Vital: Heart Rate | Hour 5:
99
Patient 2 | Vital: Systolic BP | Hour 0:
130
Patient 2 | Vital: Systolic BP | Hour 1:
135
Patient 2 | Vital: Systolic BP | Hour 2:
142
Patient 2 | Vital: Systolic BP | Hour 3:
145
Patient 2 | Vital: Systolic BP | Hour 4:
140
Patient 2 | Vital: Systolic BP | Hour 5:
133
Patient 2 | Vital: Diastolic BP | Hour 0:
85
Patient 2 | Vital: Diastolic BP | Hour 1:
88
Patient 2 | Vital: Diastolic BP | Hour 2:
92
Patient 2 | Vital: Diastolic BP | Hour 3:
95
Patient 2 | Vital: Diastolic BP | Hour 4:
90
```





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```
Patient 2 | Vital: Diastolic BP | Hour 5:
87
Patient 2 | Vital: Temperature | Hour 0:
37.2 C
Patient 2 | Vital: Temperature | Hour 1:
37.4 C
Patient 2 | Vital: Temperature | Hour 2:
37.6 C
Patient 2 | Vital: Temperature | Hour 3:
37.8 C
Patient 2 | Vital: Temperature | Hour 4:
37.5 C
Patient 2 | Vital: Temperature | Hour 5:
37.3 C
Patient 3 | Vital: Heart Rate | Hour 0:
65
Patient 3 | Vital: Heart Rate | Hour 1:
62
Patient 3 | Vital: Heart Rate | Hour 2:
58
Patient 3 | Vital: Heart Rate | Hour 3:
55
Patient 3 | Vital: Heart Rate | Hour 4:
60
Patient 3 | Vital: Heart Rate | Hour 5:
63
Patient 3 | Vital: Systolic BP | Hour 0:
105
Patient 3 | Vital: Systolic BP | Hour 1:
```





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```
Patient 3 | Vital: Systolic BP | Hour 1:
102
Patient 3 | Vital: Systolic BP | Hour 2:
98
Patient 3 | Vital: Systolic BP | Hour 3:
95
Patient 3 | Vital: Systolic BP | Hour 4:
100
Patient 3 | Vital: Systolic BP | Hour 5:
103
Patient 3 | Vital: Diastolic BP | Hour 0:
70
Patient 3 | Vital: Diastolic BP | Hour 1:
68
Patient 3 | Vital: Diastolic BP | Hour 2:
65
Patient 3 | Vital: Diastolic BP | Hour 3:
62
Patient 3 | Vital: Diastolic BP | Hour 4:
66
Patient 3 | Vital: Diastolic BP | Hour 5:
69
Patient 3 | Vital: Temperature | Hour 0:
35.8 C
Patient 3 | Vital: Temperature | Hour 1:
35.9 C
Patient 3 | Vital: Temperature | Hour 2:
36.0 C
Patient 3 | Vital: Temperature | Hour 3:
```





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Patient 3 | Vital: Temperature | Hour 5:
35.9 C

--- TODO 2: Average of Each Vital Sign, Per
Patient ---

Patient 1 Averages:

Heart Rate : 75.8
Systolic BP : 119.7
Diastolic BP : 78.0
Temperature : 36.92 C

Patient 2 Averages:

Heart Rate : 101.2
Systolic BP : 137.5
Diastolic BP : 89.5
Temperature : 37.47 C

Patient 3 Averages:

Heart Rate : 60.5
Systolic BP : 100.5
Diastolic BP : 66.7
Temperature : 35.95 C

--- TODO 3: Highest Heart Rate Reading
Overall ---

Highest Heart Rate: 112 bpm (Patient 2 at
Hour 3)





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--- TODO 4: Flag Abnormal Vitals ---

[ALERT] Patient 2 | Vital: Heart Rate |
Hour 2 | Reading: 105

[ALERT] Patient 2 | Vital: Heart Rate |
Hour 3 | Reading: 112

[ALERT] Patient 2 | Vital: Heart Rate |
Hour 4 | Reading: 108

[ALERT] Patient 2 | Vital: Systolic BP |
Hour 2 | Reading: 142

[ALERT] Patient 2 | Vital: Systolic BP |
Hour 3 | Reading: 145

[ALERT] Patient 2 | Vital: Diastolic BP |
Hour 2 | Reading: 92

[ALERT] Patient 2 | Vital: Diastolic BP |
Hour 3 | Reading: 95

[ALERT] Patient 2 | Vital: Temperature |
Hour 2 | Reading: 37.6 C

[ALERT] Patient 2 | Vital: Temperature |
Hour 3 | Reading: 37.8 C

[ALERT] Patient 3 | Vital: Heart Rate |
Hour 2 | Reading: 58

[ALERT] Patient 3 | Vital: Heart Rate |
Hour 3 | Reading: 55

[ALERT] Patient 3 | Vital: Temperature |
Hour 0 | Reading: 35.8 C

[ALERT] Patient 3 | Vital: Temperature |
Hour 1 | Reading: 35.9 C

[ALERT] Patient 3 | Vital: Temperature |





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```
[ALERT] Patient 2 | Vital: Heart Rate |
      Hour 4 | Reading: 108
[ALERT] Patient 2 | Vital: Systolic BP |
      Hour 2 | Reading: 142
[ALERT] Patient 2 | Vital: Systolic BP |
      Hour 3 | Reading: 145
[ALERT] Patient 2 | Vital: Diastolic BP |
      Hour 2 | Reading: 92
[ALERT] Patient 2 | Vital: Diastolic BP |
      Hour 3 | Reading: 95
[ALERT] Patient 2 | Vital: Temperature |
      Hour 2 | Reading: 37.6 C
[ALERT] Patient 2 | Vital: Temperature |
      Hour 3 | Reading: 37.8 C
[ALERT] Patient 3 | Vital: Heart Rate |
      Hour 2 | Reading: 58
[ALERT] Patient 3 | Vital: Heart Rate |
      Hour 3 | Reading: 55
[ALERT] Patient 3 | Vital: Temperature |
      Hour 0 | Reading: 35.8 C
[ALERT] Patient 3 | Vital: Temperature |
      Hour 1 | Reading: 35.9 C
[ALERT] Patient 3 | Vital: Temperature |
      Hour 5 | Reading: 35.9 C
```

```
=== Code Execution Successful ===
```

